

# TOPSIS Method

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# What is TOPSIS?

- **TOPSIS** (*Technique for Order Preference by Similarity to Ideal Solution*)
- A multi-criteria decision-making method (**MCDM**)
- Used for ranking and selecting alternatives based on multiple criteria
- The best alternative should:
  - be as close as possible to the ideal solution,
  - be as far as possible from the anti-ideal solution
- The method was proposed by Hwang and Yoon in 1981

# Intuition of the Method

- The ideal solution  $A^+$  contains the **best attainable values** for all criteria
- The anti-ideal solution  $A^-$  contains the **worst attainable values**
- A good alternative should simultaneously:
  - minimize the distance to  $A^+$ ,
  - maximize the distance to  $A^-$
- The output of the method is a ranking of alternatives

# Steps of the TOPSIS Method

- 1 Construct the decision matrix  $X = [x_{ij}]_{n \times m}$
- 2 Normalize the criterion values
- 3 Determine the weighted matrix
- 4 Determine the ideal solution  $A^+$  and anti-ideal solution  $A^-$
- 5 Compute Euclidean distances to  $A^+$  and  $A^-$
- 6 Compute the closeness coefficient  $C_i$  and create the ranking

## Step 1 – Vector Normalization

$$r_{ij} = \frac{x_{ij}}{\sqrt{\sum_{k=1}^n x_{kj}^2}}$$

- Each column is normalized independently
- This removes the influence of measurement units
- After normalization, the criteria become comparable
- The preference direction (min/max) is taken into account only when determining  $A^+$  and  $A^-$

## Step 2 – Weighted Matrix

$$v_{ij} = w_j r_{ij}$$

- $w_j$  – weight of the  $j$ -th criterion
- The larger the weight, the more important the criterion
- Usually the following condition is assumed:

$$\sum_{j=1}^m w_j = 1$$

## Step 3 – Ideal and Anti-Ideal Solutions

For the weighted matrix  $V = [v_{ij}]$ , we define:

$$A^+ = (v_1^+, v_2^+, \dots, v_m^+), \quad A^- = (v_1^-, v_2^-, \dots, v_m^-)$$

**For a benefit criterion (to be maximized):**

$$v_j^+ = \max_i v_{ij}, \quad v_j^- = \min_i v_{ij}$$

**For a cost criterion (to be minimized):**

$$v_j^+ = \min_i v_{ij}, \quad v_j^- = \max_i v_{ij}$$

## Step 4 – Euclidean Distances

$$S_i^+ = \sqrt{\sum_{j=1}^m \left( v_{ij} - v_j^+ \right)^2}$$

$$S_i^- = \sqrt{\sum_{j=1}^m \left( v_{ij} - v_j^- \right)^2}$$

- $S_i^+$  – distance of alternative  $i$  from the ideal solution
- $S_i^-$  – distance of alternative  $i$  from the anti-ideal solution



## Step 5 – Closeness Coefficient

$$C_i = \frac{S_i^-}{S_i^+ + S_i^-}, \quad C_i \in [0, 1]$$

- The larger the value of  $C_i$ , the better the alternative
- $C_i \approx 1$  means the alternative is close to the ideal solution
- $C_i \approx 0$  means the alternative is close to the anti-ideal solution

## Example 1 – Data

|   | Price [thousand] | Fuel consumption [l/100 km] | Comfort [1–10] |
|---|------------------|-----------------------------|----------------|
| A | 20               | 7                           | 6              |
| B | 25               | 6                           | 7              |
| C | 22               | 8                           | 5              |

- Types of criteria: price (min), fuel consumption (min), comfort (max)
- Weights:  $w = (0.4; 0.3; 0.3)$

## Example 1 – Normalization

Column norms:

$$\sqrt{20^2 + 25^2 + 22^2} = \sqrt{1509} \approx 38.85$$

$$\sqrt{7^2 + 6^2 + 8^2} = \sqrt{149} \approx 12.21$$

$$\sqrt{6^2 + 7^2 + 5^2} = \sqrt{110} \approx 10.49$$

|   | Price  | Fuel consumption | Comfort |
|---|--------|------------------|---------|
| A | 0.5149 | 0.5735           | 0.5721  |
| B | 0.6436 | 0.4915           | 0.6674  |
| C | 0.5663 | 0.6554           | 0.4767  |

## Example 1 – Weighted Matrix

$$v_{ij} = w_j r_{ij}$$

|   | Price ( $\times 0.4$ ) | Fuel consumption ( $\times 0.3$ ) | Comfort ( $\times 0.3$ ) |
|---|------------------------|-----------------------------------|--------------------------|
| A | 0.2059                 | 0.1720                            | 0.1716                   |
| B | 0.2574                 | 0.1475                            | 0.2002                   |
| C | 0.2265                 | 0.1966                            | 0.1430                   |

## Example 1 – Ideal Solutions

Since:

- price and fuel consumption are **minimized**,
- comfort is **maximized**,

we obtain:

$$A^+ = (0.2059; 0.1475; 0.2002)$$

$$A^- = (0.2574; 0.1966; 0.1430)$$

## Example 1 – Distances

$$S_A^+ = \sqrt{(0.2059 - 0.2059)^2 + (0.1720 - 0.1475)^2 + (0.1716 - 0.2002)^2} = 0.0377$$

$$S_A^- = \sqrt{(0.2059 - 0.2574)^2 + (0.1720 - 0.1966)^2 + (0.1716 - 0.1430)^2} = 0.0638$$

$$S_B^+ = \sqrt{(0.2574 - 0.2059)^2 + (0.1475 - 0.1475)^2 + (0.2002 - 0.2002)^2} = 0.0515$$

$$S_B^- = \sqrt{(0.2574 - 0.2574)^2 + (0.1475 - 0.1966)^2 + (0.2002 - 0.1430)^2} = 0.0754$$

$$S_C^+ = \sqrt{(0.2265 - 0.2059)^2 + (0.1966 - 0.1475)^2 + (0.1430 - 0.2002)^2} = 0.0782$$

$$S_C^- = \sqrt{(0.2265 - 0.2574)^2 + (0.1966 - 0.1966)^2 + (0.1430 - 0.1430)^2} = 0.0309$$

## Example 1 – Result

$$C_A = \frac{0.0638}{0.0377 + 0.0638} = 0.629$$

$$C_B = \frac{0.0754}{0.0515 + 0.0754} = 0.594$$

$$C_C = \frac{0.0309}{0.0782 + 0.0309} = 0.283$$

### Ranking

A (0.629)  $\succ$  B (0.594)  $\succ$  C (0.283)

**Conclusion:** car A is the best choice.

## Example 2 – Data

|   | Price [thousand] | Distance [km] | Area [m <sup>2</sup> ] |
|---|------------------|---------------|------------------------|
| A | 300              | 10            | 60                     |
| B | 280              | 15            | 55                     |
| C | 350              | 8             | 70                     |

- Types of criteria: price (min), distance (min), area (max)
- Weights:  $w = (0.4; 0.3; 0.3)$



## Example 2 – Normalization

Column norms:

$$\sqrt{300^2 + 280^2 + 350^2} \approx 539.35$$

$$\sqrt{10^2 + 15^2 + 8^2} \approx 19.72$$

$$\sqrt{60^2 + 55^2 + 70^2} \approx 107.35$$

|   | Price  | Distance | Area   |
|---|--------|----------|--------|
| A | 0.5562 | 0.5070   | 0.5589 |
| B | 0.5191 | 0.7605   | 0.5123 |
| C | 0.6489 | 0.4056   | 0.6520 |

## Example 2 – Weighted Matrix

|   | Price ( $\times 0.4$ ) | Distance ( $\times 0.3$ ) | Area ( $\times 0.3$ ) |
|---|------------------------|---------------------------|-----------------------|
| A | 0.2225                 | 0.1521                    | 0.1677                |
| B | 0.2077                 | 0.2282                    | 0.1537                |
| C | 0.2596                 | 0.1217                    | 0.1956                |

## Example 2 – Results

$$A^+ = (0.2077; 0.1217; 0.1956), \quad A^- = (0.2596; 0.2282; 0.1537)$$

|   | $S_i^+$ | $S_i^-$ | $C_i$ | Rank |
|---|---------|---------|-------|------|
| A | 0.0439  | 0.0858  | 0.661 | 2    |
| B | 0.1144  | 0.0519  | 0.312 | 3    |
| C | 0.0519  | 0.1144  | 0.688 | 1    |

### Ranking

$$C (0.688) \succ A (0.661) \succ B (0.312)$$

## Example 3 – Data

|   | Price [u.] | Delivery time [days] | Quality [1–10] |
|---|------------|----------------------|----------------|
| A | 100        | 5                    | 7              |
| B | 120        | 4                    | 9              |
| C | 90         | 6                    | 6              |

- Types of criteria: price (min), delivery time (min), quality (max)
- Weights:  $w = (0.3; 0.3; 0.4)$

## Example 3 – Normalization

Column norms:

$$\sqrt{100^2 + 120^2 + 90^2} \approx 180.28$$

$$\sqrt{5^2 + 4^2 + 6^2} \approx 8.77$$

$$\sqrt{7^2 + 9^2 + 6^2} \approx 12.88$$

|   | Price  | Time   | Quality |
|---|--------|--------|---------|
| A | 0.5547 | 0.5698 | 0.5433  |
| B | 0.6656 | 0.4558 | 0.6985  |
| C | 0.4992 | 0.6838 | 0.4657  |

## Example 3 – Weighted Matrix

|   | Price ( $\times 0.3$ ) | Time ( $\times 0.3$ ) | Quality ( $\times 0.4$ ) |
|---|------------------------|-----------------------|--------------------------|
| A | 0.1664                 | 0.1709                | 0.2173                   |
| B | 0.1997                 | 0.1368                | 0.2794                   |
| C | 0.1498                 | 0.2051                | 0.1863                   |

## Example 3 – Results

$$A^+ = (0.1498; 0.1368; 0.2794), \quad A^- = (0.1997; 0.2051; 0.1863)$$

|   | $S_i^+$ | $S_i^-$ | $C_i$ | Rank |
|---|---------|---------|-------|------|
| A | 0.0728  | 0.0569  | 0.439 | 2    |
| B | 0.0499  | 0.1155  | 0.698 | 1    |
| C | 0.1155  | 0.0499  | 0.302 | 3    |

### Ranking

$$B (0.698) \succ A (0.439) \succ C (0.302)$$

# Interpretation of Results

- The values  $C_i$  allow us to compare the quality of alternatives
- The larger the value of  $C_i$ , the better the position in the ranking
- Differences between  $C_i$  values show how much the alternatives differ from one another
- TOPSIS provides not only the winner, but a complete ordering of variants



# Applications of the TOPSIS Method

- selection of a product, technology, or supplier,
- logistics and supply chain management,
- investment analysis,
- risk assessment,
- recommendation systems,
- evaluation of projects, employees, and offers

# Advantages and Disadvantages

## Advantages

- simple and intuitive geometric interpretation,
- easy implementation,
- ability to include many criteria and weights,
- simultaneous consideration of the ideal and anti-ideal solutions

## Disadvantages

- the result depends on the chosen weights,
- sensitivity to the normalization method,
- no consideration of dependencies between criteria,
- the possibility of *rank reversal*

- TOPSIS is one of the most popular MCDM methods
- It is based on the distance from the ideal and anti-ideal solutions
- It allows us to build a clear ranking of alternatives
- Key elements are: choice of weights, normalization, and correct interpretation of criteria

# Assignment – Function/Module Input

- Write your own implementation of the TOPSIS method
- Solve your own decision problem
- You may use **any programming language**
- The input to your function/module should be:
  - the **decision matrix**  $X = [x_{ij}]_{n \times m}$ ,
  - the **weight vector**  $w = (w_1, w_2, \dots, w_m)$ ,
  - the **criterion type vector** indicating whether each criterion is **min** or **max**
- Use at least **10 alternatives**
- Use at least **3 criteria**
- Use **exactly the same normalization** as in this presentation:

$$r_{ij} = \frac{x_{ij}}{\sqrt{\sum_{k=1}^n x_{kj}^2}}$$

- Do not use a ready-made TOPSIS library
- Show the **data sources** used for all criteria and alternatives

# Assignment – Function/Module Output

- The output of your function/module should be:
  - the **preference vector**

$$C = (C_1, C_2, \dots, C_n),$$

where  $C_i$  is the preference point (closeness coefficient) for alternative  $i$ ,

- the **ranking vector** defining the final order of alternatives
- Present the full TOPSIS procedure for your problem:
  - normalized matrix,
  - weighted matrix,
  - ideal solution,
  - anti-ideal solution,
  - distances from the ideal and anti-ideal solutions,
  - preference points,
  - final ranking
- Briefly interpret the results and explain which alternative is the best